

REPORT

Title

Tic-Tac-Toe Game Using Python

Aim

To develop a Python program to implement a two-player Tic-Tac-Toe game and determine the winner or draw.

Problem Statement

Tic-Tac-Toe is a two-player game played on a 3×3 grid. Each player takes turns placing either X or O in an empty cell. A player wins when three of their symbols are placed consecutively in a row, column, or diagonal. If all cells are filled without a winner, the game ends in a draw.

Requirements

- Python 3.x
- Python IDLE, VS Code, or any Python IDE
- Basic knowledge of Python functions, loops, lists, and conditional statements

Procedure

1. Create a 3×3 game board.
2. Initialize all board positions as empty.

3. Assign X to the first player and O to the second player.
4. Display the current board.
5. Get the position selected by the current player.
6. Check whether the selected position is valid and empty.
7. Place the player's symbol in the selected position.
8. Check all possible winning combinations.
9. If a winning combination is found, declare the player as the winner.
10. If all nine positions are filled without a winner, declare a draw.
11. Otherwise, switch the player and continue the game.

Program

```
board = [" "] * 9

def display_board():
    print()
    print(board[0], "|", board[1], "|", board[2])
    print("---+---+---")
    print(board[3], "|", board[4], "|", board[5])
    print("---+---+---")
    print(board[6], "|", board[7], "|", board[8])
    print()

def check_winner(player):
    winning_positions = [
        (0,1,2), (3,4,5), (6,7,8),
        (0,3,6), (1,4,7), (2,5,8),
        (0,4,8), (2,4,6)
    ]

    for a, b, c in winning_positions:
        if board[a] == board[b] == board[c] == player:
            return True
    return False

def game():
    player = "X"
```

```

for turn in range(9):
    display_board()

    position = int(input(
        f"Player {player}, enter position (1-9): "
    ))

    while position < 1 or position > 9 or board[position-1]
!= " ":
        print("Invalid position. Try again.")
        position = int(input("Enter position (1-9): "))

    board[position-1] = player

    if check_winner(player):
        display_board()
        print(f"Player {player} wins!")
        return

    player = "O" if player == "X" else "X"

display_board()
print("The game is a draw!")

game()

```

Sample Output

```

X | X | X
---+---+---
  | O |
---+---+---
  | O |

Player X wins!

```

Conclusion

The Tic-Tac-Toe game was successfully implemented using Python. The program allows two players to make alternate moves, validates the moves, checks all possible winning combinations, and identifies whether a player wins or the game ends in a draw.

